ALGORITHMS - PART II SECOND-ORDER METHODS

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Outline

1. Newton method for unconstrained problems
2. Newton method for equality-constraint problems
3. Log-barrier method for inequality-constrained problems

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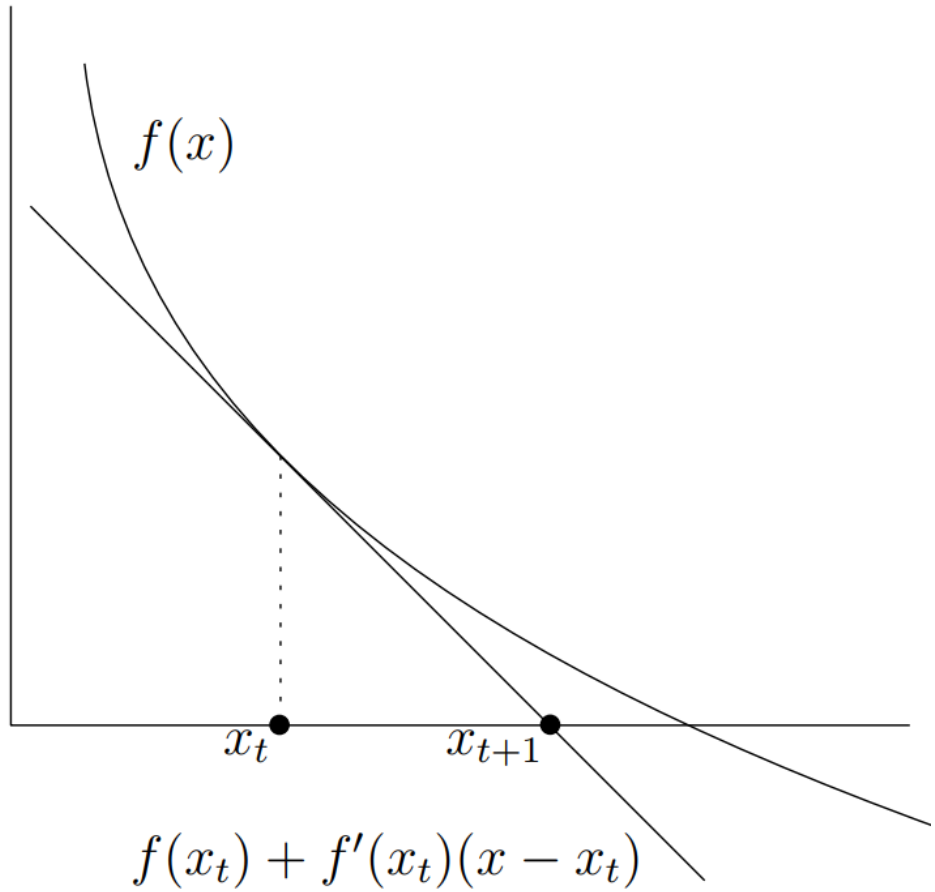
1. Newton method for unconstrained problems
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3. Log-barrier method for inequality-constrained problems

Motivation: Newton-Raphson method for finding a root

Find a zero of differentiable function $f : \mathbb{R} \to \mathbb{R}$,

$$x_{t+1} := x_t - \frac{f(x_t)}{f'(x_t)},$$

x_{t+1} solves
$f(x_t) + f'(x_t)(x - x_t) = 0.$



Unconstrained minimization

Given unconstrained, smooth convex optimization

$$\min_x f(x),$$

where $f(x)$ is convex, twice differentiable, and $dom(f) = \mathbb{R}^n$.
Newton method repeats

$$x^{(k)} = x^{(k-1)} - (\nabla^2 f(x^{(k-1)}))^{-1} \nabla f(x^{(k-1)}).$$

Recall: gradient descent

Recall that gradient descent chooses initial $x^{(0)} \in \mathbb{R}^n$, and update $x^{(k)}$ which minimize a quadratic approximation of $f(x^{(k-1)})$:

$$f(y) \approx f(x^{(k-1)}) + \nabla f(x^{(k-1)})^T (y - x) + \frac{1}{2t} \|y - x^{(k-1)}\|_2^2,$$

and

$$x^{(k)} = arg \min_y f(x^{(k-1)}) + \nabla f(x^{(k-1)})^T (y - x) + \frac{1}{2t} \|y - x^{(k-1)}\|_2^2.$$

Newton's method

Newton method uses a "better" quadratic approximation:

$$f(y) \approx f(x^{(k-1)}) \nabla f(x^{(k-1)})^T (y - x) + \\ + \frac{1}{2}(y - x^{(k-1)})^T \nabla^2 f(x^{(k-1)})(y - x^{(k-1)}),$$

and minimizes over y to yield the update rule

$$x^{(k)} = arg \min_y f(x^{(k-1)}) + \nabla^T f(x^{(k-1)})(y - x) \\ + \frac{1}{2}(y - x^{(k-1)})^T \nabla^2 f(x^{(k-1)})(y - x^{(k-1)}).$$

Note that Newton method does not use step size t.

Example Newton's method vs gradient descent

We consider the problem of minimizing

$$f(x_1, x_2) = \frac{10x_1^2 + x_2^2}{2} + 5\log(1 + e^{-x_1 - x_2}).$$

Gradient and Hessian is given by:

$$\nabla f(x_1, x_2) = \begin{bmatrix} 10x_1 - \frac{5e^{-x_1-x_2}}{1+e^{-x_1-x_2}} \\ x_2 - \frac{5e^{-x_1-x_2}}{1+e^{-x_1-x_2}} \end{bmatrix}$$

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 10 + \frac{5e^{-x_1-x_2}}{(1+e^{-x_1-x_2})^2} & \frac{5e^{-x_1-x_2}}{(1+e^{-x_1-x_2})^2} \\ \frac{5e^{-x_1-x_2}}{(1+e^{-x_1-x_2})^2} & 1 + \frac{5e^{-x_1-x_2}}{(1+e^{-x_1-x_2})^2} \end{bmatrix}$$

Comparison between GD and Newton methods

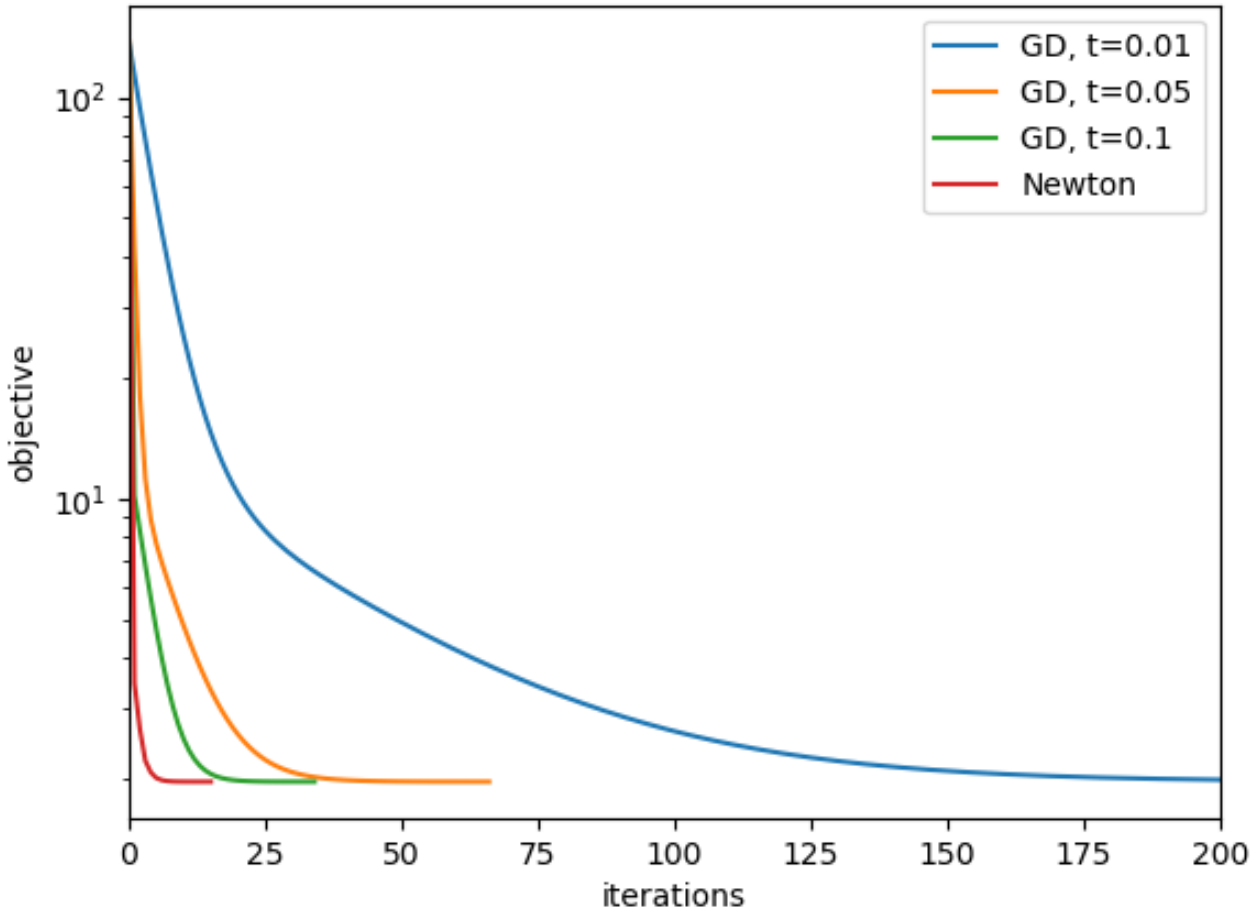


Figure: Comparison between GD and Newton methods. The stopping condition is $\|x(k) - x(k-1)\| < 10^{-6}$.

Newton method with stepsize

The pure Newton method doesn't use step size. We can apply Newton method with step size:

$$x^{(k)} = x^{(k-1)} - t_k(\nabla^2 f(x^{(k-1)}))^{-1}\nabla f(x^{(k-1)}),$$

in which t_k is calculated by backtracking line search.

Affine Invariance of Newton method

Important property Newton's method: **affine invariance**.
Affine transformation $y = Ax$, where $A \in \mathbb{R}^{n \times n}$ and invertible. Let $g(x) = f(Ax)$, hence,

$$\nabla g(x) = A^T \nabla f(y); \qquad \nabla^2 g(x) = A^T \nabla^2 f(y) A;$$

Newton update for minimizing $g(x)$ is:

$$\begin{aligned} x^{(k)} &= x^{(k-1)} - (\nabla^2 g(x^{(k-1)}))^{-1} \nabla g(x^{(k-1)}) \\ &= x^{(k-1)} - A^{-1} (\nabla^2 f(y))^{-1} (A^T)^{-1} A^T \nabla f(y) \\ &= x^{(k-1)} - A^{-1} (\nabla^2 f(y))^{-1} \nabla f(y) \end{aligned}$$

(Hint. $(AB)^{-1} = B^{-1} A^{-1}$ và $(A^T)^{-1} = (A^{-1})^T$.)
Hence,

$$Ax^{(k)} = Ax^{(k-1)} - (\nabla^2 f(y))^{-1} \nabla f(y),$$

which is the Newton update minimizing $f(Ax)$.
Affine Invariance property does not hold for gradient descent method.

Comparison to first-order methods

At a high-level:

- **Memory**: each iteration of Newton's method requires $O(n^2)$ storage ($n \times n$ Hessian); each gradient iteration requires $O(n)$ storage (n-dimensional gradient)
- **Computation**: each Newton iteration requires $O(n^3)$ flops (solving a dense $n \times n$ linear system); each gradient iteration requires $O(n)$ flops (scaling/adding n-dimensional vectors)
- **Backtracking**: backtracking line search has roughly the same cost, both use $O(n)$ flops per inner backtracking step
- **Conditioning**: Newton's method is not affected by a problem's conditioning, but gradient descent can seriously degrade

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Equality-constrained problem

Consider now a problem with equality constraints, as in

$$\min_x f(x) \quad \text{subject to} \quad Ax = b$$

- f convex, continuously differentiable
- $A \in \mathbb{R}^{p \times n}$ with rank $A = p$.

optimality conditions: $x^\star$ is optimal if and only if there exists a $v^\star$ such that

$$\nabla f(x^\star) + A^T v^\star = 0, \quad Ax^\star = b$$

Equality Constrained Quadratic Minimization

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^T Px + q^T x + r \\ \text{subject to} & Ax = b, \end{array}$$

where $P \in \mathbb{S}_+^n$

Optimality condition:

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \nu^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

- The coefficient matrix is called the KKT matrix
- KKT matrix is nonsingular **if and only if**

$$Ax = 0,\ x \neq 0 \quad \Longrightarrow \quad x^T Px > 0$$

- Equivalent condition for nonsingularity:

$$P + A^T A > 0 \quad \text{(positive definite)}$$

Equality-constrained Newton method

In equality-constrained Newton's method, we start with $x^{(0)}$ such that $Ax^{(0)} = b$. Then we repeat the updates

$$x^+ = x + tv,$$

where

$$v = \arg\min_{Az=0} \nabla f(x)^T (z - x) + \frac{1}{2}(z - x)^T \nabla^2 f(x)(z - x)$$

This keeps x^+ in feasible set, since $Ax^+ = Ax + tAv = b + 0 = b$ Furthermore, v is the solution to minimizing a quadratic subject to equality constraints. We know from KKT conditions that v satisfies

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

for some w. Hence Newton direction v is again given by solving a linear system in the Hessian (albeit a bigger one)

Quasi-Newton methods

If the Hessian is too expensive (or singular), then a quasi-Newton method can be used to approximate $\nabla^2 f(x)$ with $H > 0$, and we update according to

$$x^+ = x - tH^{-1}\nabla f(x)$$

- Approximate Hessian H is recomputed at each step. Goal is to make H^{-1} cheap to apply (possibly, cheap storage too)
- Convergence is fast: superlinear, but not the same as Newton. Roughly n steps of quasi-Newton make same progress as one Newton step
- Very wide variety of quasi-Newton methods; common theme is to "propogate" computation of H across iterations

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Log barrier function

Consider the convex optimization problem

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to } & h_i(x) \leq 0, i = 1, \ldots, m \\ & Ax = b \end{aligned}$$

We assume that $f, h_1, \ldots, h_m$ are convex, twice differentiable, each with domain $\mathbb{R}^n$. The function

$$\phi(x) = -\sum_{i=1}^{m} \log(-h_i(x))$$

is called the log barrier for the above problem.
Its domain is the set of strictly feasible points,
$\{x : h_i(x) < 0, i = 1, \ldots, m\}$, which is assumed nonempty. (Note: Slater's condition, strong duality holds)

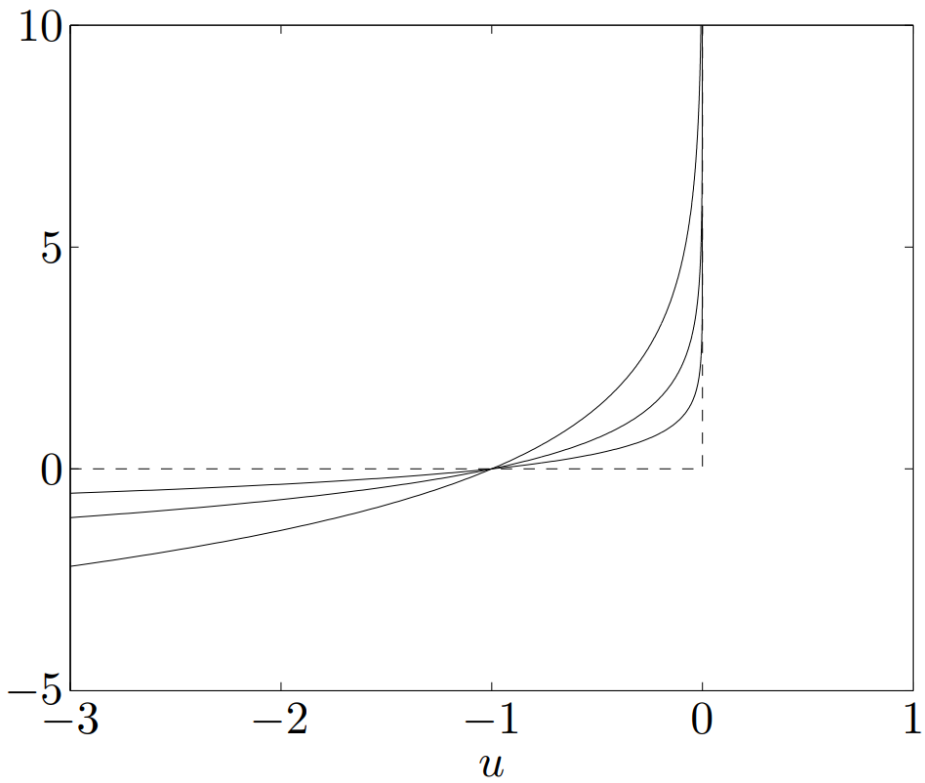
Ignoring equality constraints for now, our problem can be written as

$$\min_x f(x) + \sum_{i=1}^{m} I_{\{h_i(x) \leq 0\}}(x)$$

We can approximate the sum of indicators by the log barrier:

$$\min_x f(x) - \frac{1}{t} \sum_{i=1}^{m} \log(-h_i(x))$$

where $t > 0$ is a large number
This approximation is more accurate for larger t. But for any value of t, the log barrier approaches ∞ if any $h_i(x) \to 0$



Log barrier calculus

For the log barrier function

$$\phi(x) = -\sum_{i=1}^{m} \log(-h_i(x))$$

we have for its gradient:

$$\nabla\phi(x) = -\sum_{i=1}^{m} \frac{1}{h_i(x)} \nabla h_i(x)$$

and for its Hessian:

$$\nabla^2\phi(x) = \sum_{i=1}^{m} \frac{1}{h_i(x)^2} \nabla h_i(x) \nabla h_i(x)^T - \sum_{i=1}^{m} \frac{1}{h_i(x)} \nabla^2 h_i(x)$$

Central path

Consider barrier problem:

$$\min_x \; tf(x) + \phi(x)$$
$$\text{subject to } Ax = b$$

The **central path** is defined by solution $x^\star(t)$ with respect to $t > 0$

- Hope is that, as $t \to \infty$, we will have $x^\star(t) \to x^\star$, solution to our original problem
- Why don't we just set t to be some huge value, and solve the above problem? Directly seek solution at **end** of central path?
- Problem is that this is seriously inefficient in practice
- Much more efficient to **traverse** the central path, as we will see

Central path for LP

An important special case: barrier problem for a linear program

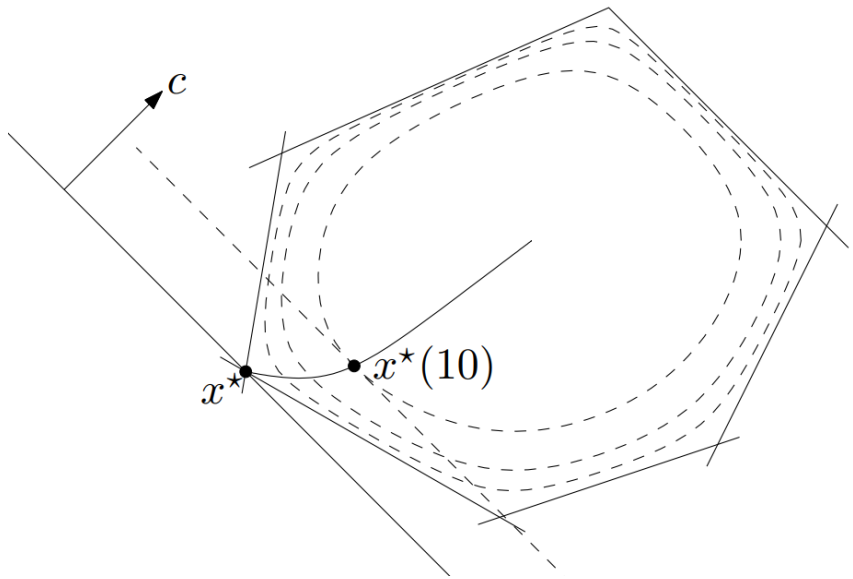
$$\min_x \; tc^T x - \sum_{i=1}^{m} \log(e_i - d_i^T x)$$

The barrier function corresponds to polyhedral constraint $Dx \leq e$

Gradient optimality condition:

$$0 = tc - \sum_{i=1}^{m} \frac{1}{e_i - d_i^T x^\star(t)} d_i$$

This means that gradient $\nabla\phi(x^\star(t))$ must be parallel to $-c$, i.e., hyperplane $\{x : c^T x = c^T x^\star(t)\}$ lies tangent to contour of ϕ at $x^\star(t)$



KKT conditions and duality

Central path is characterized by its KKT conditions:

$$t\nabla f(x^\star(t)) - \sum_{i=1}^{m} \frac{1}{h_i(x^\star(t))}\nabla h_i(x^\star(t)) + A^T w = 0,$$

$$Ax^\star(t) = b, \quad h_i(x^\star(t)) < 0, \quad i = 1, \ldots, m$$

for some $w \in \mathbb{R}^m$.
From central path points, we can derive feasible dual points for our original problem. Given $x^\star(t)$ and corresponding w, we define

$$u_i^\star(t) = -\frac{1}{th_i(x^\star(t))}, \quad i = 1, \ldots, m, \quad v^\star(t) = w/t$$

We claim $u^\star(t), v^\star(t)$ are dual feasible for original problem, whose Lagrangian is

$$L(x, u, v) = f(x) + \sum_{i=1}^{m} u_i h_i(x) + v^T(Ax - b)$$

Why?

- Note that $u_i^\star(t) > 0$ since $h_i(x^\star(t)) < 0$ for all $i = 1, \ldots, m$
- Further, the point $(u^\star(t), v^\star(t))$ lies in domain of Lagrange dual function $g(u, v)$, since by definition

$$\nabla f(x^\star(t)) + \sum_{i=1}^{m} u_i(x^\star(t)) \nabla h_i(x^\star(t)) + A^T v^\star(t) = 0$$

That is, $x^\star(t)$ minimizes Lagrangian $L(x, u^\star(t), v^\star(t))$ over x, so $g(u^\star(t), v^\star(t)) > -\infty$

Duality gap

This allows us to bound suboptimality of $f(x^\star(t))$, with respect to original problem, via the **duality gap**. We compute

$$\begin{aligned} g(u^\star(t), v^\star(t)) &= f(x^\star(t)) + \sum_{i=1}^{m} u_i^\star(t) h_i(x^\star(t)) + v^\star(t)^T (Ax^\star(t) - b) \\ &= f(x^\star(t)) - m/t \end{aligned}$$

That is, we know that $f(x^\star(t)) - f^\star \leq m/t$
This will be very useful as a stopping criterion; it also confirms the hope that $x^\star(t) \to x^\star$ as $t \to \infty$

Perturbed KKT conditions

We can now reinterpret central path $(x^\star(t), u^\star(t), v^\star(t))$ as solving the perturbed KKT conditions:

$$\nabla f(x) + \sum_{i=1}^{m} u_i \nabla h_i(x) + A^T v = 0$$

$$u_i \cdot h_i(x) = -1/t, \quad i = 1, \ldots, m$$

$$h_i(x) \leq 0, \quad i = 1, \ldots, m, \quad Ax = b$$

$$u_i \geq 0, \quad i = 1, \ldots, m$$

Only difference between these and actual KKT conditions for our original problem is second line: these are replaced by

$$u_i \cdot h_i(x) = 0, \quad i = 1, \ldots, m$$

i.e., **complementary slackness**, in actual KKT conditions

Barrier method

The **barrier method** solves a sequence of problems

$$\min_x \; tf(x) + \phi(x)$$
$$\text{subject to } Ax = b$$

for increasing values of $t > 0$, until duality gap satisfies $m/t \leq \epsilon$.
We fix $t^{(0)} > 0$, $\mu > 1$. We use Newton to compute $x^{(0)} = x^\star(t)$, solution to barrier problem at $t = t^{(0)}$. For $k = 1, 2, 3, \ldots$

- Solve the barrier problem at $t = t^{(k)}$, using Newton initialized at $x^{(k-1)}$, to yield $x^{(k)} = x^\star(t)$
- Stop if $m/t \leq \epsilon$, else update $t^{(k+1)} = \mu t$

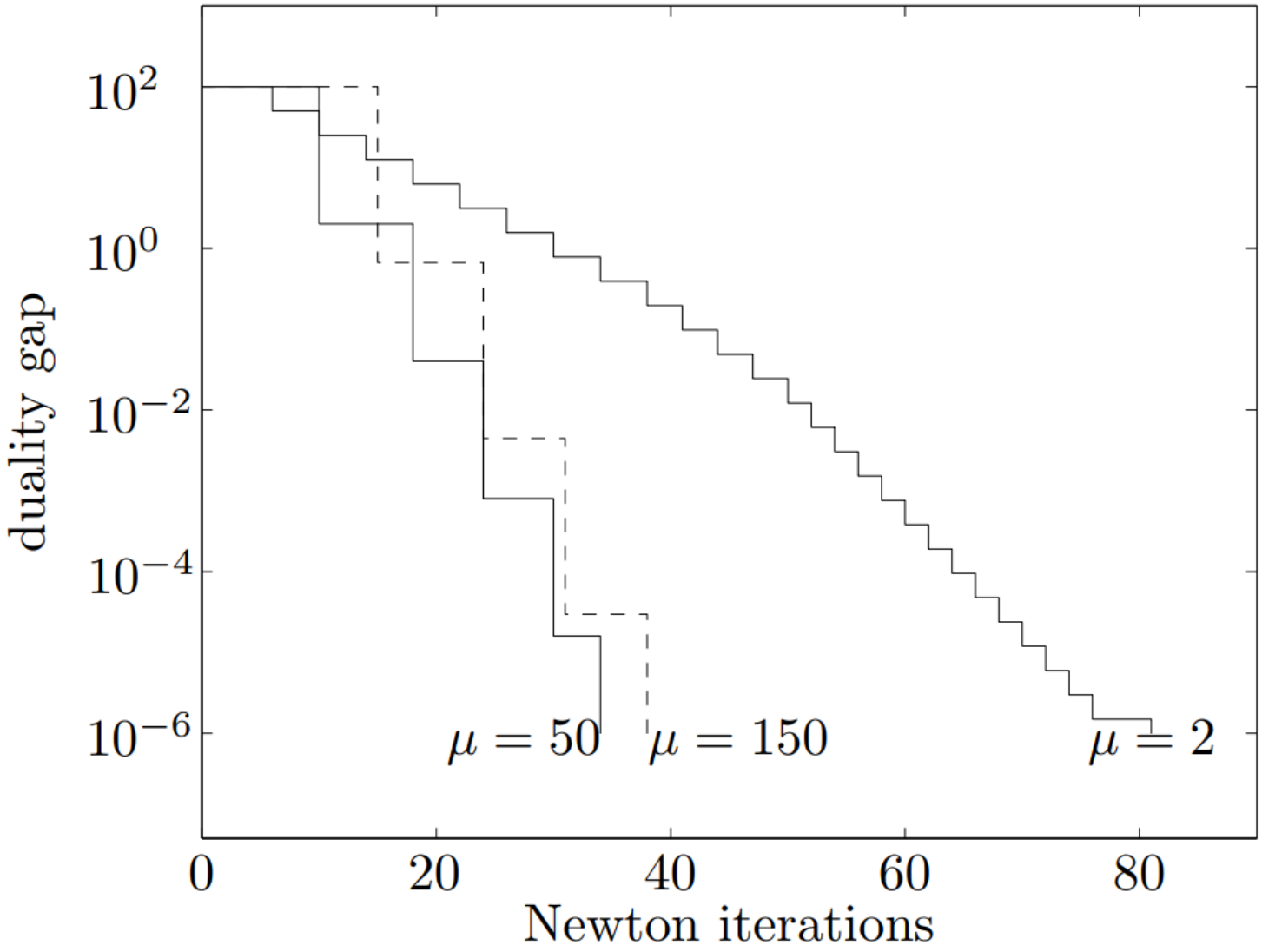
The first step above is called a centering step (since it brings $x^{(k)}$ onto the central path)

Considerations

- Choice of μ: if μ is too small, then many outer iterations might be needed; if μ is too big, then Newton's method (each centering step) might take many iterations
- Choice of $t^{(0)}$: if $t^{(0)}$ is too small, then many outer iterations might be needed; if $t^{(0)}$ is too big, then the first Newton solve (first centering step) might require many iterations

Fortunately, the performance of the barrier method is often quite robust to the choice of μ and $t^{(0)}$ in practice

Example of a small LP In $n = 50$ dimensions, $m = 100$ inequality constraints (from B & V page 571):



Convergence analysis

Assume that we solve the centering steps exactly. The following result is immediate

Theorem: The barrier method after k centering steps satisfies

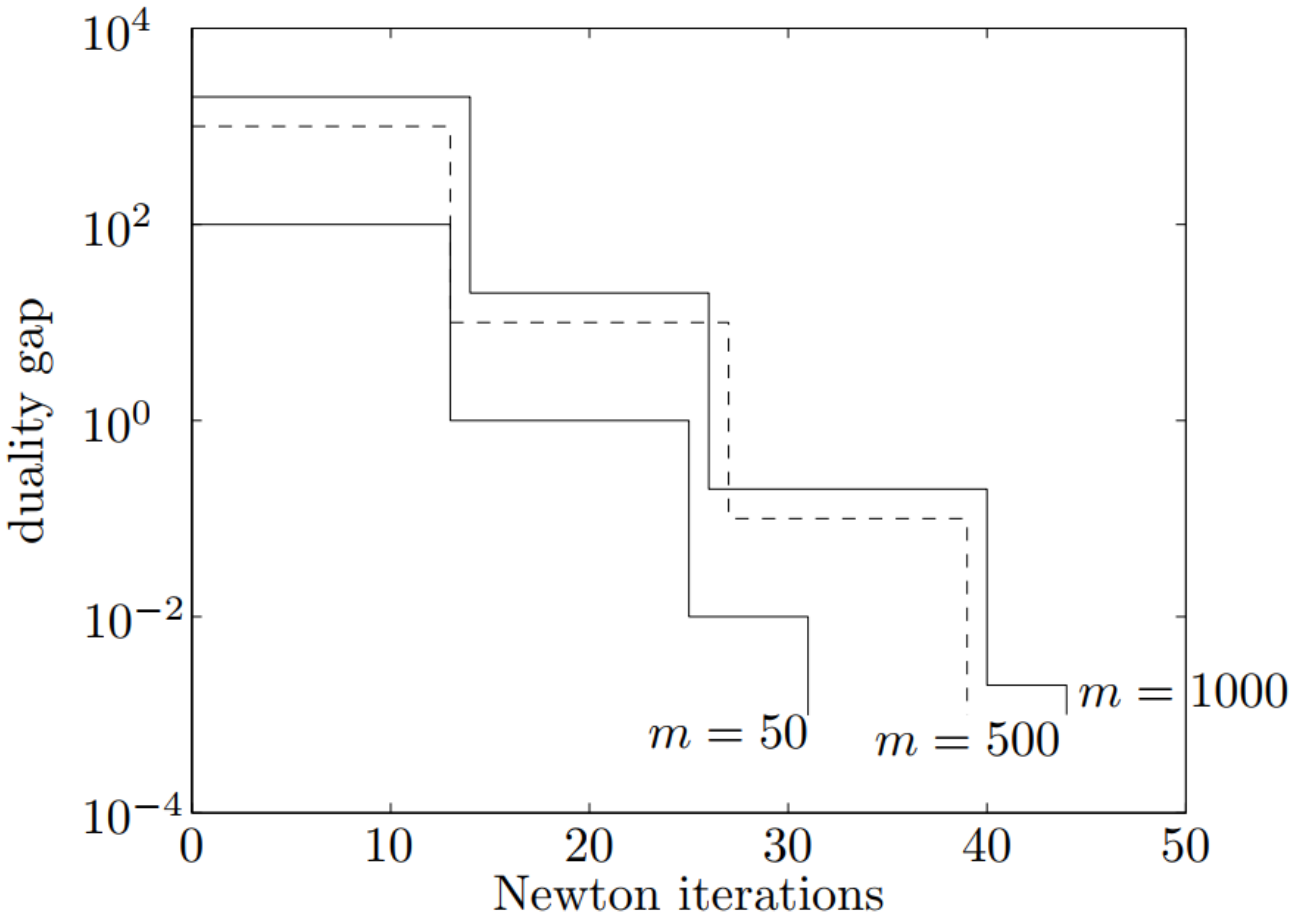
$$f(x^{(k)}) - f^\star \leq \frac{m}{\mu^k t^{(0)}}$$

In other words, to reach a desired accuracy level of ϵ, we require

$$\frac{\log(m/(t^{(0)}\epsilon))}{\log \mu}$$

centering steps with the barrier method (plus initial centering step)

Example of barrier method progress for an LP with m constraints (From B & V page 575):



Can see roughly linear convergence in each case, and logarithmic scaling with m

How many Newton iterations?

Informally, due to careful central path traversal, in each centering step, Newton is already in quadratic convergence phase, so takes nearly constant number of iterations

This can be formalized under self-concordance. Suppose:

- The function $tf + \phi$ is self-concordant
- Our original problem has bounded sublevel sets

Then we can terminate each Newton solve at appropriate accuracy, and the total number of Newton iterations is still $O(\log(m/(t^{(0)}\epsilon))$ (where constants do not depend on problem-specific conditioning). See Chapter 11.5 of B & V

Importantly, $tf + \phi = tf - \sum_{i=1}^{m} \log(-h_i)$ is self-concordant when f, h_i are all linear or quadratic. This covers all LPs, QPs, QCQPs

References

- Convex optimization book by Stephen Boyd
- Convex Optimization course at CMU